

Yosys: Your Solution for Synthesis

PRIYA RAVINDRAN

So, you are working on your circuit design and your Verilog code is ready! You have tried your hand at simulation and the code is functioning correctly. What next? Synthesis is the answer.

Carrying forward from last month's DVD story on simulation with Icarus Verilog, take that code and synthesise it with Yosys. Read on to find out what Yosys offers for synthesis of your code.

Yosys is an open source tool for performing logical synthesis. Falling under Internet Software Consortium (ISC) licence category, it is a free software that takes in your Verilog/Very high speed integrated circuit Hardware Description Language (VHDL) code and gives out a gate-level netlist that your printed circuit board (PCB) designing software understands.

How it helps

Yosys is built to let a regular user move through the stages of chip designing in an easy, hassle-free manner. With presence on [Reddit](#), [Stack Overflow](#) and [GitHub](#) (github.com/cliffordwolf/yosys), the tool can run on Ubuntu, Fedora and Windows operating systems (OSes). Here are a few features that would keep you thinking about the experience, long after you have finished working on your design.

With most synthesis tools being proprietary, Yosys is one tool you can easily download and start working with your code anytime you want. To work with Yosys, you will just need a C++ compiler with C++11 support and a few tools like GNU Flex, GNU Bison and GNU Make.

Yosys reads your Verilog code, supporting Verilog 2005 version, con-

Are you a beginner? Here is what you need

Hardware design generally follows a standard pattern: design entry, simulation, synthesis, layout, verification and fabrication. Icarus Verilog is a tool that aids the front-end of your design by letting you synthesise your code.

Design entry. Describe your design in words or using schematics. The word description is generally in Verilog, VHDL or other such HDLs.

Simulation. Check the functionality of your design. Simulate how your design would work under different conditions by assigning values to inputs in your design, via a testbench. Observe the outputs as display or waveform.

Synthesis. Convert your Verilog code into a machine-understandable format called the netlist, which is simply a list of all wires and connections in your design. The netlist is what is carried forward from the front-end into the layout stage, where the design is converted into a pictorial format, which is then printed. This stage is technology-specific and may also include analysis for timing and power factors.

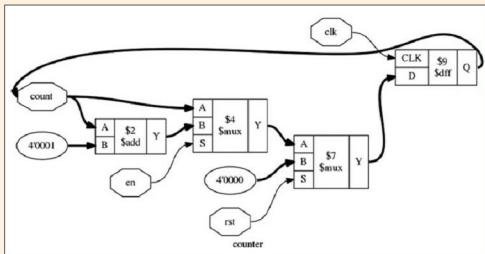


Fig. 1: View of an RTL netlist

verts it into its internal format, elaborates the design to establish design hierarchy, performs optimisations and converts the code into a netlist.

Standard algorithms. There are a variety of algorithms, called passes, to perform register transfer level (RTL) behavioural and logic synthesis of your code. These are modelled to suit different applications, and you can modify these to suit your requirements using synthesis scripts.

A synthesis script is simply a file with commands that will be executed in sequence on the file being read by the tool. The table (next page) shows a basic synthesis script that can be used to synthesise most designs. You

can even add additional passes to existing ones by extending Yosys C++ code base. This is further made easy by making sure all passes use the same interface, so as to ease the effort from the user to understand or modify the source code. Yosys even supports multi-bit values and multi-port memories.

Flexibility. You can choose to generate three kinds of netlists, namely, a simple RTL netlist that describes the circuit in terms of registers and data transfer, a complementary metal oxide semiconductor (CMOS) gate-level netlist that converts a circuit to its basic form characterised by CMOS gates and